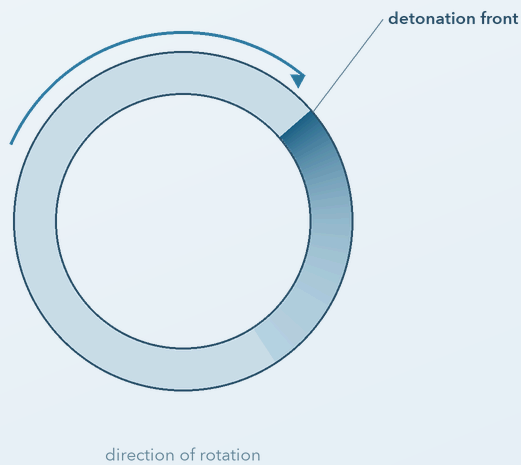


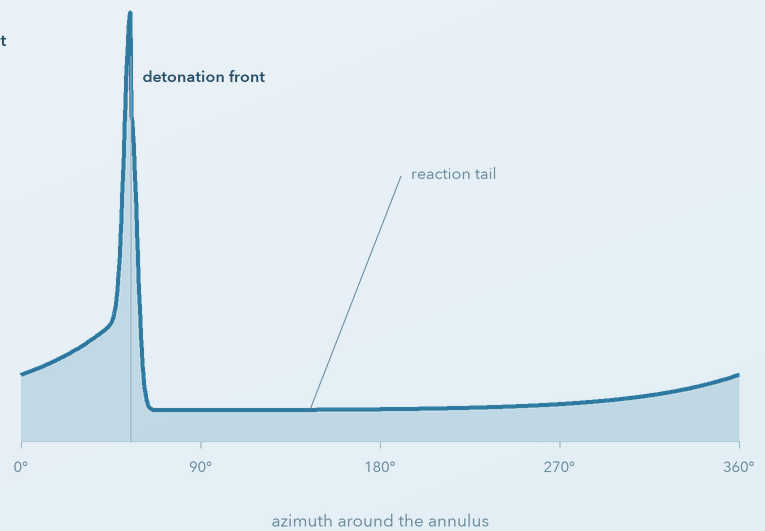
Resonance

A student-led liquid-methane rotating detonation rocket engine at UIUC.

Annular pressure field



Pressure around one revolution

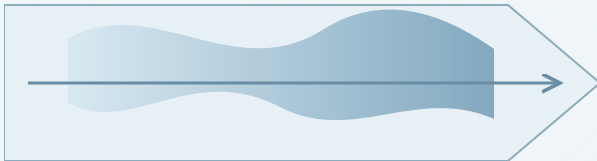


Resonance Mk1 is designed to demonstrate sustained rotating detonation, calibrated thrust, and thermal survival at UIUC's SPIRIT facility. It is the first step toward compact upper-stage and in-space propulsion.

Why detonation changes the engine

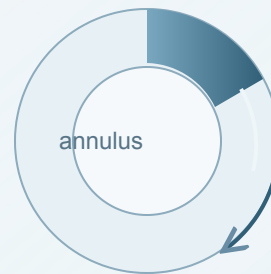
A conventional rocket burns through a steady flame. An RDRE uses a shock-driven flame that races around an annular chamber thousands of times each second.

Steady combustion



A flame front advances through a longer chamber.

Rotating detonation



The shock compresses and burns fresh propellant.

15-25%

higher efficiency potential

Up to 40%

shorter chamber

About 10×

power density

These are published field-level advantages, not measured Resonance performance. Resonance is being built to test that promise in a compact liquid-methane engine.

Project lead



Syed Aabis Akhtar

Aerospace Engineering, UIUC
Engine Fluids, Liquid Rocketry at Illinois

Aabis has built products and technical systems since high school, then moved into propulsion at UIUC. At Liquid Rocketry at Illinois, he has built feed-system hardware from P&IDs and supported pressure checks, cold flows, ignition testing, and Overture Mk1 hotfire operations.

Built early

Co-founded StellarSOL in tenth grade and raised \$50,000 in equity-free grants and prizes.

Learned fast

Learned Swift and built AsymmED in four days without owning a Mac; won Apple's 2023 Swift Student Challenge.

Shipped systems

Built Solathon to 150,000+ downloads, then moved into feed-system hardware and hotfire operations.



Overture Mk1 hotfire, May 2026. Aabis supported fluids and test-stand operations.

Resonance Mk1 architecture

Resonance Mk1 is a pressure-fed, water-cooled, predetonated liquid-oxygen and liquid-methane RDRE. The injector forms a fresh methane-oxygen layer; the rotating front consumes it, and the burned gas exits through the nozzle.

2.0 kN

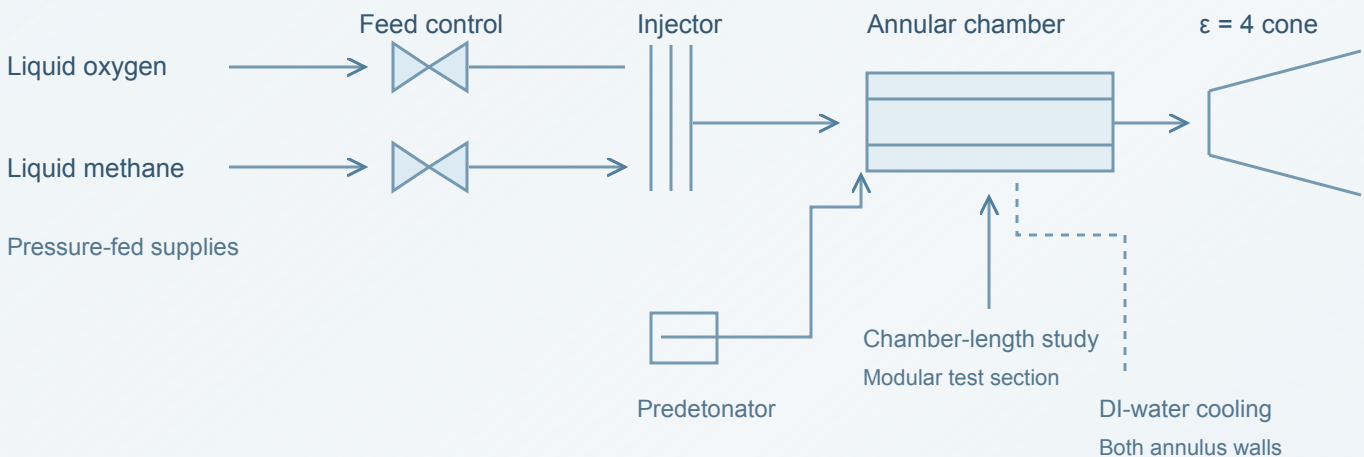
Nominal; 1.5-2.5 kN operating envelope

350 psia

Nominal chamber pressure; 250 psia commissioning

85 × 5 mm

Annular gas path



Modular hot section

One injector head supports 38, 51, and 64 mm face-to-throat lengths. The 51 mm barrel fires first; the other lengths build on the same interface.

Build approach

LPBF GRCo-42 with computed-tomography inspection is preferred if a qualified supplier is available. A CNC OFHC calorimeter is the backup route.

Resonance Mk1 at a glance

A compact, pressure-fed engine sized for an instrumented first firing at SPIRIT.

2.0 kN

Nominal thrust

350 psia

Chamber pressure

LOX / CH₄

Propellants

≥ 5 s

First success; 10 s objective

85 × 5 mm

Annular gas path

0.874 kg/s

Nominal total flow at O/F 3.00

Combustor

Three cooled barrels at 1.5, 2.0, and 2.5 inches. The 2.0-inch barrel fires first.

Cooling

Deionised water on the outer wall and centerbody for multi-second operation.

Ignition

An independently qualified predetonator initiates the Mk1 engine.

Nozzle

Expansion ratio 4 for early tests; the final throat topology remains in review.

First success

A rotating-detonation signature, calibrated thrust, and at least five seconds of thermal survival.

Why Resonance is different

Resonance starts with the propellants, cooling, and measurements needed for a useful engine program, rather than treating detonation as a one-second demonstration.

Liquid methane from the start

The first article uses the clean, reusable upper-stage propellant pair the program is intended to advance.

Actively cooled for real data

Both annulus walls are water-cooled so the engine can run for at least five seconds and be inspected afterward.

Proof beyond a bright plume

Dynamic pressure identifies the rotating mode; calibrated thrust and flow show what the engine actually delivers.

Built around iteration

A replaceable injector and modular hot section let the team change hardware without rebuilding the entire engine.

$\approx 2 \text{ km/s}$

detonation-front speed

$\approx 0.13 \text{ ms}$

one lap around the annulus

4 channels

planned dynamic-pressure stations

Testing at UIUC

SPIRIT

The liquid-propulsion test facility at the University of Illinois Urbana-Champaign.

Resonance uses an existing university test cell and occupies a modest fraction of its stated envelope.

10 kN

Maximum thrust

The 2 kN engine target is one-fifth of the stated limit.



500 psi

Maximum chamber pressure

The 350 psia design point remains below the stated chamber limit.

12 in

Maximum diameter

A 5.0-5.5 inch engine leaves room for cooling, plumbing, and instrumentation.



SPIRIT keeps sponsor support focused on the engine, cooling system, diagnostics, and test hardware rather than a new test site.

From design review to first firing

The first approved hotfire remains targeted for Spring 2027.

Oct-Nov
2026

Review and release

Close the injector, cooling, joints, diagnostics, and test interfaces before releasing hardware.

Dec-Feb
2027

Build and inspect

Manufacture the 51 mm hot section, injector, predetonator, fixtures, and cooling hardware.

Mar-Apr
2027

Cold flow and checkout

Verify valves, flow, cooling, sensors, calibration, aborts, and the stand interface.

Spring 2027

Commission and fire

Begin near 250 psia, around 1.4 kN on the current sizing, then expand toward the 350 psia design point from measured data.

**Success is
measured**

Show a rotating wave after the starter shuts off, record calibrated thrust and flow, and keep the hot section intact for at least five seconds.

Build Resonance with us

SPIRIT provides the test cell. Sponsor support turns the reviewed design into inspected hardware and measurements we can trust.

Cooled copper hardware

GRCop-42 printing or machined copper, finish machining, and CT inspection of every cooling passage.

Injector and cold flow

CNC and microdrilling, flow fixtures, metering, and a repeatable acceptance test.

Fast diagnostics

Four dynamic-pressure channels at 250 kS/s or faster per channel, plus calibration support.

Cooling and operations

Water-loop hardware, valves, fittings, consumables, seals, spares, and test-day support.

Support a complete work package

Cash, manufacturing credit, sensors, inspection, materials, and engineering review can each move one part from design to verified hardware.

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